Two-Minute Interview Narration

A clear and honest explanation of the ICFDR malnutrition classification project.

Project: Child Malnutrition Status Classification (ICFDR ML Project)

Source archive: ICFDR---ML-Project-main.zip | Main notebook: MalnutritionDetect.ipynb | Additional notebooks: SMOTE.ipynb and Oversampling on the whole dataset.ipynb

Narration (approximately two minutes)

“This project is a supervised machine-learning classification problem. The objective is to classify a record into one of four malnutrition-related categories: Stunting, Overweight, Underweight or Wasting, using features such as sex, age, height and weight, along with income-group indicators in the original data.”

“The dataset contains 837 rows and eight columns, with no missing values. The major challenge is class imbalance: 696 records are Stunting, 94 Overweight, 43 Underweight and only four Wasting. That means a model can achieve more than 80 percent accuracy by mostly predicting Stunting, while failing to identify the rare classes. So I focus on per-class precision and recall, macro-F1 and confusion matrices rather than accuracy alone.”

“The notebooks explore Random Forest and Support Vector Machines, along with a small neural network. Random Forest is useful for nonlinear feature interactions and does not require scaling. SVM is sensitive to feature scales, so the SMOTE notebook standardizes features before tuning C and the kernel with GridSearchCV. The project also experiments with SMOTE to balance the training data. The correct workflow is to split first and apply SMOTE only to the training set.”

“The main notebook reports 72.22 percent test accuracy for one Random Forest run and 84.13 percent for a linear SVM run. However, the test partition contains no Wasting examples, and the minority-class F1 scores are poor. There is also a separate notebook that applies SMOTE before splitting the full dataset, which causes leakage and makes its evaluation invalid. The neural-network branch also uses a binary sigmoid output for a four-class problem, so it needs to be reformulated.”

“My conclusion is that the project is a useful learning exercise in multiclass classification and imbalance handling, but the results are not sufficient for real-world nutritional screening. I would collect more minority-class data, use stratified leakage-free cross-validation, compare class weighting with SMOTE, and obtain domain-expert validation before considering any practical use.”

Likely interview questions

1. Why is this multiclass classification? The target has four mutually exclusive categorical labels.

2. Why is accuracy misleading? The 83.2% majority class means an always-Stunting classifier scores 83.2% but detects no minority classes.

3. What does SMOTE do? It interpolates between minority samples and their nearest minority neighbours to create synthetic training examples.

4. Why must SMOTE happen after splitting? If it happens before splitting, synthetic points can be influenced by observations that later appear in validation/test, leaking information.

5. Why scale for SVM? SVM relies on distances/margins; large-scale features can dominate smaller-scale features.

6. Why Random Forest? It captures nonlinear interactions, is relatively robust to feature scaling and provides feature-importance diagnostics.

7. What is macro-F1? It calculates F1 for each class and averages them equally, so rare classes are not drowned out by the majority.

8. Why is the test result incomplete? The test set contains zero Wasting examples, so Wasting performance cannot be estimated.

9. What is wrong with the neural network? A single sigmoid output and binary cross-entropy are for binary classification, but the task has four classes.

10. Which model is best? The saved metrics do not establish a trustworthy winner; the SVM has higher accuracy, but minority-class performance is weak and the test split is incomplete.

11. What would you improve first? More representative rare-class data, leakage-free stratified validation, and evaluation using macro-F1/per-class recall.

12. Can it be used for clinical decisions? No. It is a small educational dataset and requires clinical/domain validation, external testing and human oversight.